

SE Lab

(UE24CS341A)

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1. Objective and Project Continuity

The objective of Lab 2 is to convert the functional requirements identified in Lab 1 into an Agile backlog, organize the work into Epics and User Stories, prioritize and estimate the stories using Fibonacci-style Story Points, create Scrum sprints in Jira, simulate progress through To Do → In Progress → Done, and analyse the resulting Scrum reports.

The selected problem is the Blood Bank Inventory & Emergency Donor Matcher. The system focuses on emergency blood-request handling, blood inventory management, compatibility verification, donor matching/notification, donor availability and expiry management.

2. Agile Epics

ID	Epic	Description	Lab 1 Traceability
EPIC-01	Emergency Blood Request & Triage	Enable verified medical centres/emergency requesters to submit complete, prioritized emergency blood requests.	FR-004
EPIC-02	Blood Inventory & Expiry Management	Maintain accurate blood inventory and identify blood units approaching expiry.	FR-003, FR-005
EPIC-03	Emergency Donor Matching & Notification	Cross-reference emergency demand with inventory, verify donor compatibility and notify eligible donors.	FR-001, FR-002

3. User Story Backlog, Priority and Story Points

Key	Epic	Story	User Story	Priority	Points
SCRUM-8	EPIC-01	Submit Emergency Blood Request	As a verified medical centre, I want to submit an emergency blood request with quantity, urgency level, and patient blood profile, so that the blood bank can triage and process the request correctly.	High	5
SCRUM-9	EPIC-01	Validate Requester and Mandatory Fields	As the system, I want to validate requester verification and mandatory request fields, so that incomplete or unauthorized emergency requests are rejected.	High	3
SCRUM-10	EPIC-02	Update Blood Inventory	As a Blood Bank Manager, I want to add, remove, and adjust inventory counts by blood group and component type, so that current stock levels are accurate.	High	5
SCRUM-11	EPIC-02	Flag Expiring Blood Units	As a Blood Bank Manager, I want the system to flag blood units expiring within 48 hours, so that I can take action and reduce wastage.	Medium	3
SCRUM-12	EPIC-03	Check Inventory for Emergency Availability	As a Blood Bank Manager, I want emergency requests checked against real-time inventory, so that available stock is used before donor matching is initiated.	High	3
SCRUM-13	EPIC-03	Verify ABO and Rh(D) Compatibility	As the system, I want to verify ABO and Rh(D) compatibility for each candidate donor, so that incompatible donors are never notified.	High	5
SCRUM-14	EPIC-03	Match and Notify Eligible Donors	As a Blood Bank Manager, I want the system to match eligible donors within the required radius and send SMS alerts, so that emergency shortages can be addressed quickly.	High	5
SCRUM-15	EPIC-03	Notify Donors for Emergency Requests	As the system, I want compatible donors to receive emergency-request notifications, so that urgent blood requirements can be communicated quickly.	High	3
SCRUM-16	EPIC-03	Confirm Donor Availability	As a donor, I want to confirm my availability after receiving an emergency notification, so that the blood bank knows which donors can respond.	Medium	3

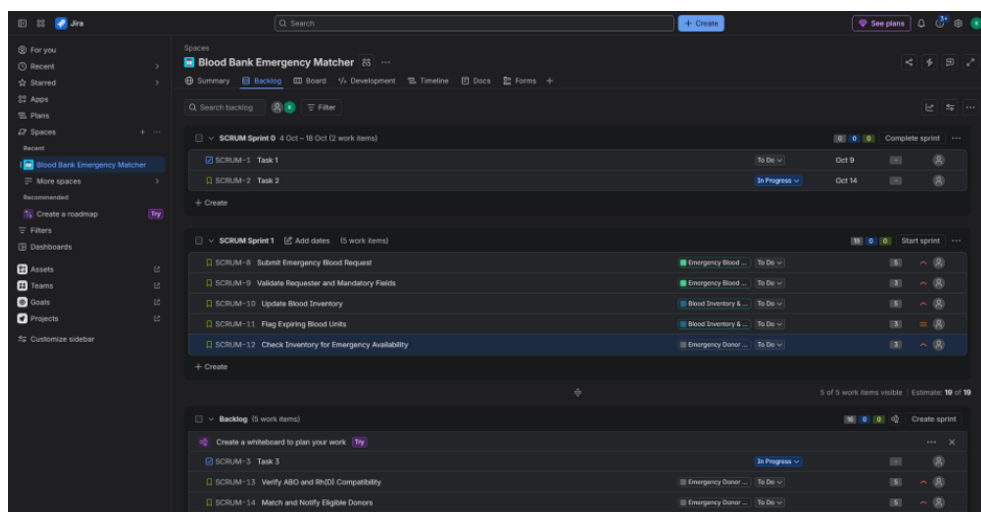
4. Estimation Approach

Story Points were assigned using the Fibonacci-style scale recommended for the exercise. The estimates represent relative effort, complexity, uncertainty and dependencies rather than hours.

Story-point value	Interpretation
3	Moderate complexity with limited dependencies and a relatively predictable implementation.
5	Higher complexity involving multiple validation, data-handling, workflow or integration concerns.

5. Actual Jira Backlog Evidence

The Jira backlog was populated with the emergency blood request, validation, inventory, expiry, availability, compatibility and donor-matching stories. The captured workspace also contains an additional reporting task, SCRUM-3, which was later handled separately from the nine core user stories.



6. Sprint Execution

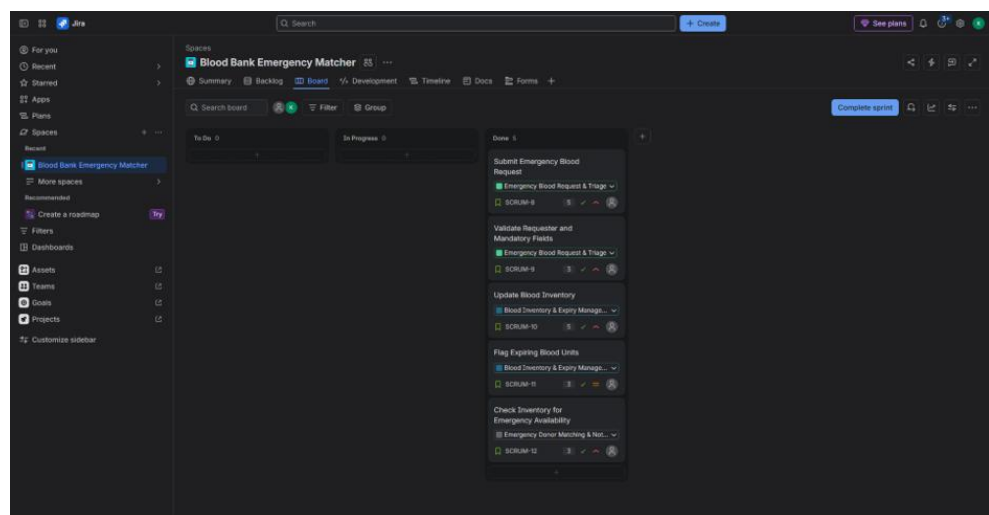
The Jira workspace was used to simulate Scrum execution. The two main sprint groups used in the exercise were Sprint 1 and Sprint 2. A later Sprint 3 contains an additional reporting task.

6.1. Sprint 1

Sprint goal: Implement and validate emergency blood request and inventory management capabilities.

Key	Story	Priority	Points
SCRUM-8	Submit Emergency Blood Request	High	5
SCRUM-9	Validate Requester and Mandatory Fields	High	3
SCRUM-10	Update Blood Inventory	High	5
SCRUM-11	Flag Expiring Blood Units	Medium	3
SCRUM-12	Check Inventory for Emergency Availability	High	3

Total Sprint 1 scope: 19 story points.

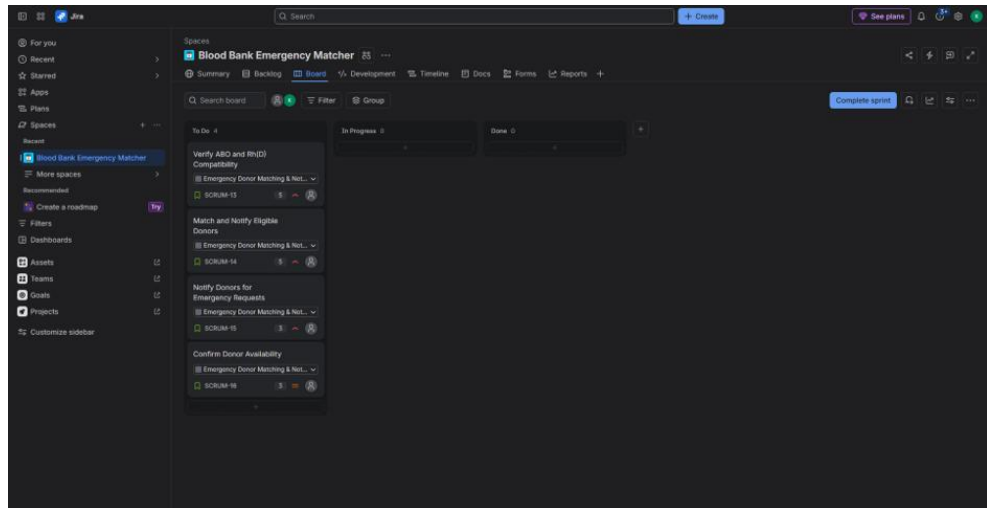


6.2. Sprint 2

Sprint goal: Complete donor compatibility, matching, notification and donor-availability capabilities.

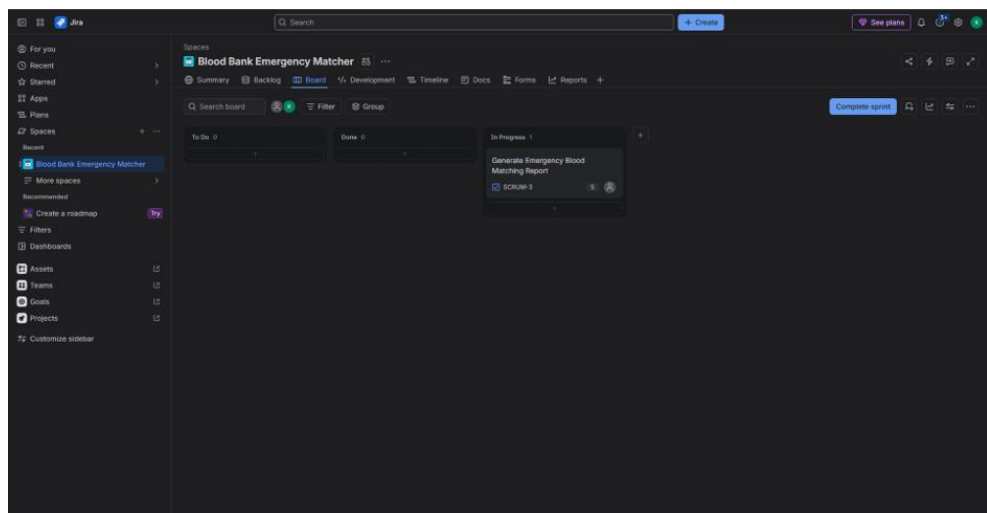
Key	Story	Priority	Points
SCRUM-13	Verify ABO and Rh(D) Compatibility	High	5
SCRUM-14	Match and Notify Eligible Donors	High	5
SCRUM-15	Notify Donors for Emergency Requests	High	3
SCRUM-16	Confirm Donor Availability	Medium	3

Total Sprint 2 scope: 16 story points.



6.3. Sprint 3 Task

The Jira workspace later contains SCRUM-3 - Generate Emergency Blood Matching Report, estimated at 5 story points and shown In Progress. This is treated as an additional task rather than one of the nine core user stories used for the Lab 2 sprint plan.

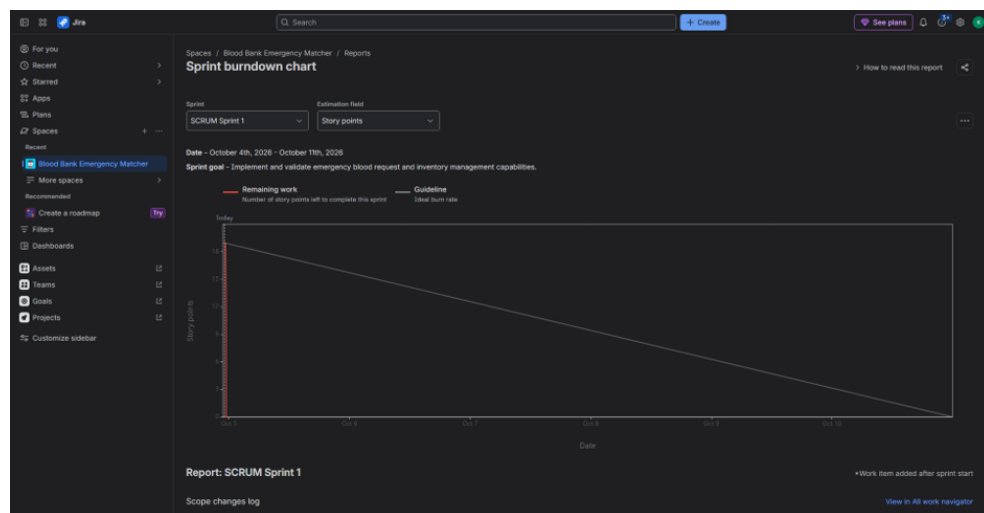


7. Scrum Reports and Analysis

The following evidence was captured from Jira Reports after the sprint activities. The screenshots are retained as submission evidence. Where Jira reports depend on external development/deployment integrations, the report is described exactly as displayed rather than inventing missing data.

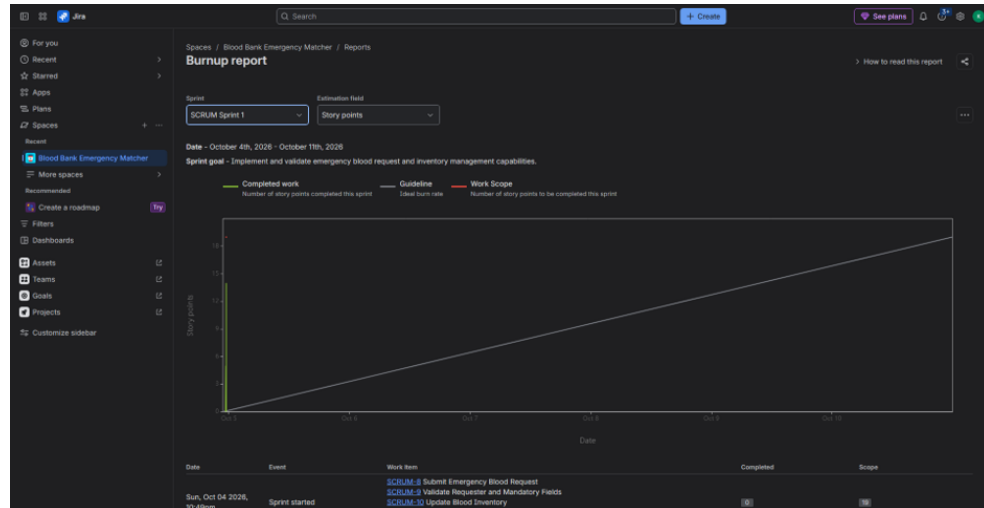
7.1. Sprint Burndown - Sprint 1

The Sprint 1 burndown report shows a 19-point scope and an ideal guideline from the beginning to the end of the one-week sprint. The captured remaining-work trace is concentrated at the beginning of the report, while the Sprint board provides the direct evidence of story status.



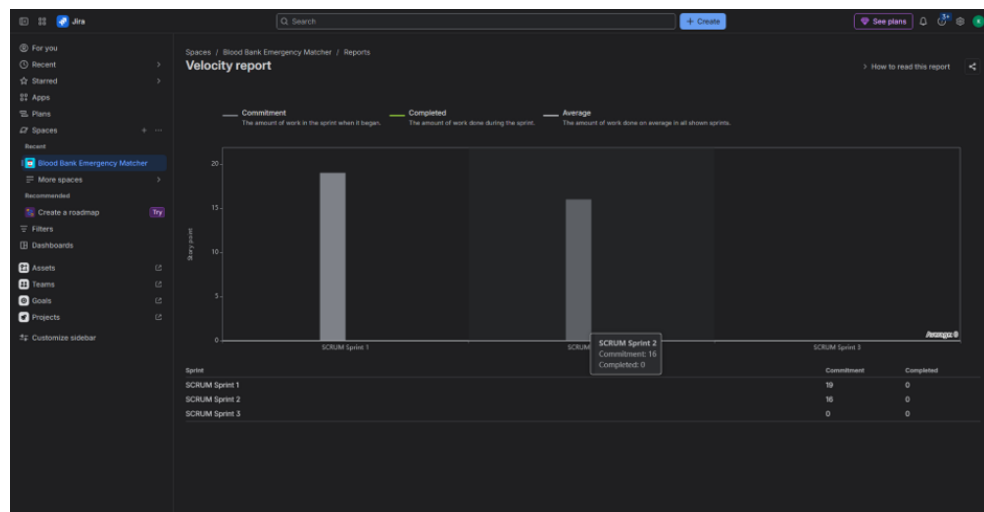
7.2. Burnup Report - Sprint 1

The burnup report compares completed work with the sprint's work scope. The captured report shows a 19-point scope and records the sprint start and story events used to build the chart.



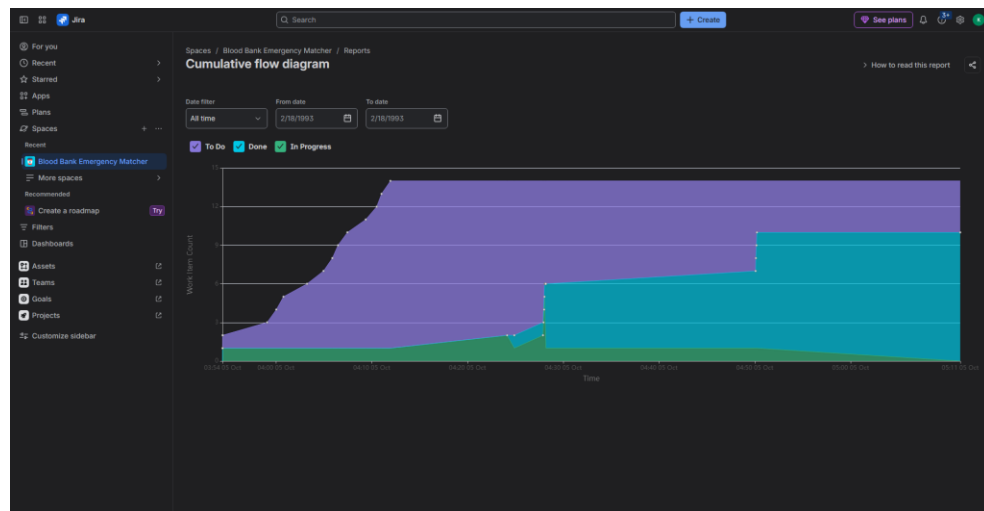
7.3. Velocity Report

The velocity report records the amount of work committed at the start of each sprint and the amount Jira currently counts as completed for reporting purposes. At capture time it displayed commitments of 19 points for Sprint 1, 16 points for Sprint 2 and 0 for Sprint 3; the displayed completed values were 0. This reflects the report state at the time of capture and is not replaced with inferred values.



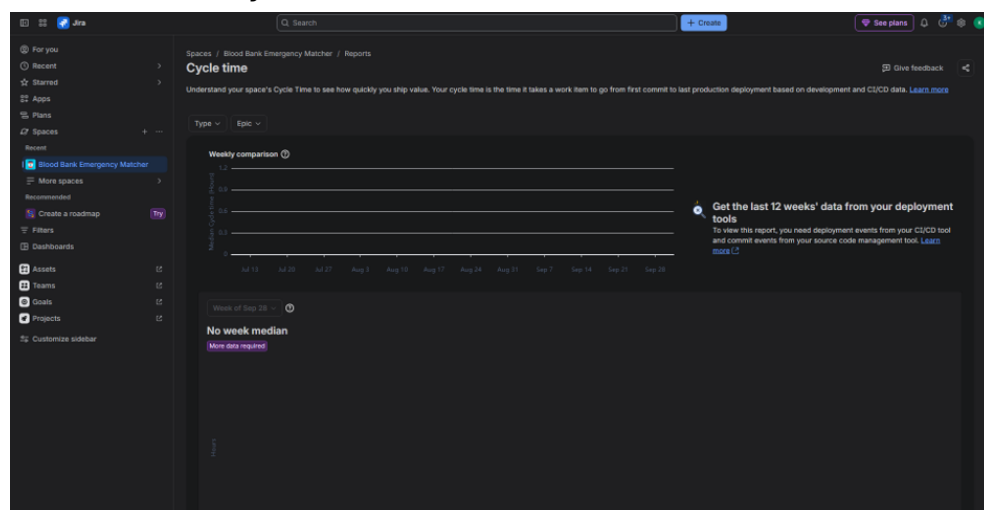
7.4. Cumulative Flow Diagram

The cumulative flow diagram visualizes how work items accumulated across To Do, In Progress and Done over time. It provides a visual view of workflow movement and can be used to identify periods of increasing work-in-progress.



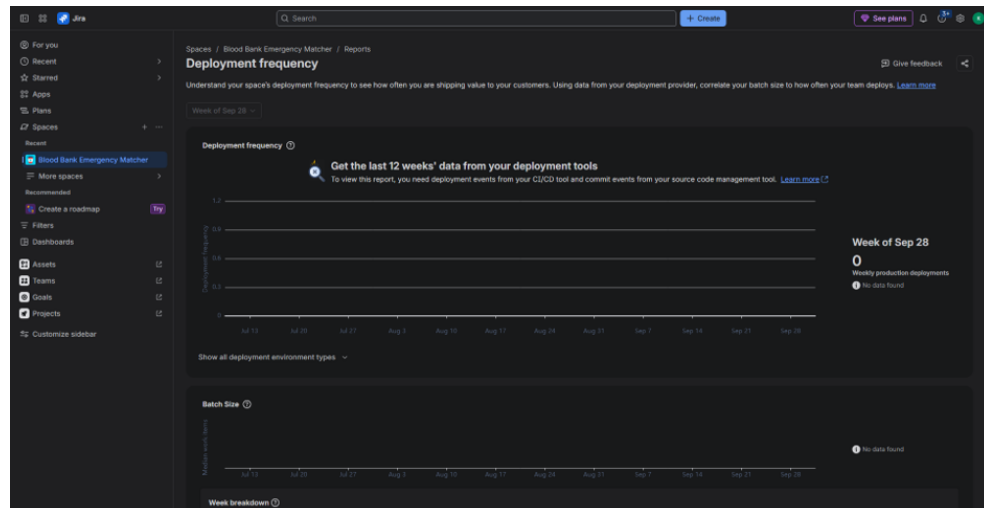
7.5. Cycle Time Report

The Cycle Time report displayed 'No week median' and 'More data required'. Jira also indicated that deployment and source-control events are required for the extended cycle-time analysis. Therefore, no cycle-time value is invented in this submission.



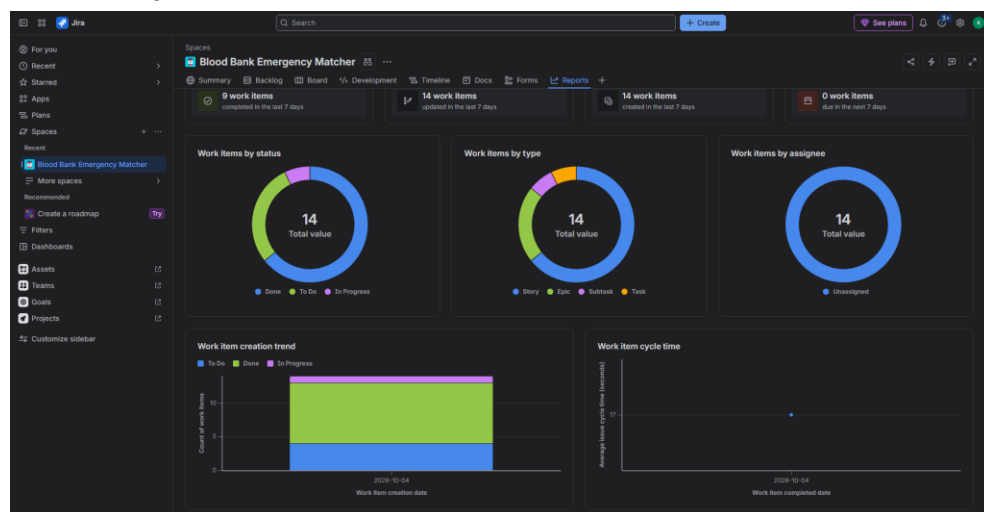
7.6. Deployment Frequency Report

The Deployment Frequency report displayed 0 weekly production deployments and 'No data found'. Jira requested deployment events from a CI/CD tool and commit events from source-code management. Since these integrations were not connected for this exercise, the report contains no deployment history.



7.7. Jira Summary / Reports Dashboard

The Jira Reports overview captured 14 work items in the project dashboard, including 5 shown as completed, 14 updated and 14 created in the displayed reporting window, with 0 due in the next seven days.



8. Reflection and Analysis

8.1. Did the estimations reflect the actual effort?

The Fibonacci estimates provided a useful relative measure of complexity. The 3-point stories represented moderate workflow or validation tasks, while the 5-point stories represented larger units involving multiple steps, data handling, compatibility checks or notification logic. The Jira execution demonstrates that the stories could be organized into manageable sprint scopes of 19 and 16 points.

8.2. Was the backlog well prioritized?

Yes. High-priority work was concentrated around emergency request intake, validation, inventory accuracy, availability checking, compatibility verification and donor matching/notification. These features are directly connected to the emergency workflow and therefore provide greater business and safety value than supporting features.

8.3. How did the simulated sprints align with the plan?

Sprint 1 focused on the core emergency request and inventory workflow. Sprint 2 moved to donor compatibility, matching, notification and availability. This sequence follows the logical dependency of the system: a valid request must be established and checked against inventory before compatible donors are matched and contacted.

8.4. What insights did the burndown chart provide?

The burndown chart provides the intended comparison between remaining story points and the ideal burn-rate guideline. The captured chart establishes the 19-point Sprint 1 scope. In this workspace, the board status and report data are not perfectly synchronized at every capture point, so the report should be interpreted together with the Sprint board rather than using the chart alone as evidence of completion.

8.5. What did the additional reports reveal?

Velocity provided a historical view of sprint commitment and reported completion. The cumulative flow diagram showed workflow-state movement. Cycle Time and Deployment Frequency could not provide meaningful numerical metrics because the Jira project did not have enough development/deployment integration data. This is an important practical lesson: Agile reporting quality depends on consistent issue-state updates as well as connected development and CI/CD tooling.

9. Conclusion

The Blood Bank Emergency Matcher was successfully modeled as an Agile Scrum project in Jira. The Lab 1 requirements were transformed into Epics and User Stories, priorities and story points were assigned, sprint scopes were organized, and Jira's Scrum board and reporting features were used to simulate and analyse project progress. The exercise also demonstrated an important distinction between issue-management reports and engineering delivery metrics: reports such as Cycle Time and Deployment Frequency require sufficient connected development and deployment data.